

AI Pitching & Negotiation Coach for Startup Fundraising

Project Proposal

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Introduction and Background

Effective communication is the cornerstone of success in startup fundraising, venture capital pitching, and accelerator programs. However, many professionals lack access to high-quality training that is both realistic and scalable. Traditional approaches—such as workshops and seminars—often fail to replicate the pressure, unpredictability, and complexity of real-world interactions.

This project proposes an AI-powered negotiation coach tailored to startup fundraising, venture capital pitching, and accelerator programs. Leveraging large language models (LLMs), natural language processing (NLP), and voice interaction systems, our platform simulates lifelike negotiation scenarios. Users engage in dynamic conversations with AI counterparts that adapt to their responses in real time.

Key differentiator:

The system delivers personalized, actionable feedback based on behavioral science, providing tactical improvements in phrasing, emotional tone, and strategy—beyond generic tips. This makes it an indispensable tool for professionals aiming to master high-stakes communication without the risks of real-world trial and error.

Problem Description and Project Objective

Professionals in startup fundraising, venture capital pitching, and accelerator programs often face high-stakes negotiations where poor communication can lead to lost opportunities. Existing training solutions lack adaptive personalization, realistic voice interaction, and continuous improvement tracking.

Project Objective:

To develop an AI-powered negotiation training platform that simulates investor meetings, pitch presentations, and funding negotiations and provides targeted, data-driven feedback to enhance persuasion, clarity, and confidence.

Application Areas and Related Work

Application Areas:

1. Startup fundraising, venture capital pitching, and accelerator programs
2. Professional development and upskilling
3. Risk-free negotiation practice environments

Related Work:

ACE: LLM-based negotiation coach with strong text-based feedback but limited real-time interactivity.

IAGO & Pactum: Agent-based negotiation tools that work well for structured scenarios but lack natural conversational flow.

Our project advances these systems by offering fully voice-driven, context-adaptive simulations.

Methodology and Scope

Methodology:

1. Dataset preparation from real-world investor meetings, pitch presentations, and funding negotiations transcripts.
2. Fine-tuning LLMs for negotiation-specific conversation flow.
3. Integration of speech-to-text (Whisper) and text-to-speech (ElevenLabs or Azure TTS).
4. Real-time feedback engine using sentiment analysis, persuasion scoring, and strategy evaluation.
5. Web platform with session recording and analytics.

Scope Limitations:

1. Focuses on verbal negotiation skills; non-verbal communication not covered initially.
2. Limited to English in Phase 1.

Tools and Technology

1. Python, JavaScript, FastAPI, Next.js, Tailwind CSS
2. PostgreSQL/Supabase for data storage
3. LangChain for prompt chaining
4. Whisper AI for STT, Azure TTS for output

5. GPU-enabled cloud hosting (24GB VRAM minimum)

Milestones

Dataset collection & preprocessing	25 Aug 2025
LLM integration & tuning	15 Sep 2025
Voice interaction pipeline	30 Sep 2025
Feedback engine implementation	20 Oct 2025
Beta testing with target audience	10 Nov 2025
Final deployment	15 Dec 2025

References:

- [1] R. Shea, A. Kallala, X. L. Liu, M. W. Morris, and Z. Yu, “ACE: A LLM-Based Negotiation Coaching System,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Miami, FL, USA, 2024, pp. 12720–12749.
- [2] A. Chatterjee, S. Miller, and N. Parepally, “AgreeMate: Training LLMs to Haggle,” *arXiv preprint arXiv:2412.18690*, December 2024.
- [3] Y. Zhou, H. He, A. W. Black, and Y. Tsvetkov, “A Dynamic Strategy Coach for Effective Negotiation,” in *Proceedings of the 20th Annual SIGdial Meeting on Discourse and Dialogue*, Stockholm, Sweden, 2019, pp. 367–378.
- [4] M. Vaccaro, M. Caoson, H. Ju, S. Aral, and J. R. Curhan, “Advancing AI Negotiations: New Theoretical Insights and Evidence from a Large-Scale Autonomous Negotiations Competition,” *arXiv preprint arXiv:2503.06416*, March 2025.
- [5] F. Bianchi, P. J. Chia, M. Yuksekgonul, J. Tagliabue, D. Jurafsky, and J. Zou, “Evaluating the Negotiation Abilities of LLMs: The NegotiationArena Platform and Analysis,” *arXiv preprint arXiv:2402.05863*, February 2024.

